

Abhay Gupta

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Portfolio: [abgup.com](#) & [notes.abgup.com](#)

Current: [Seattle, WA 98104](#)

U.S. Citizen

After taking leave to help with family, I have been able to sustainably build a product design firm and teach simultaneously. I am willing to transition back into a corporate role if the right job arises!

SKILLS

Languages	Frameworks	Software Tools	Modeling/Analysis	Hardware Tools
• Python • C# & Lua • Javascript & VBA • HTML/CSS • Bash & Powershell • Vimscript • C/C++ • LabView	• WinForms/WPF • Flask • Jekyll • Cordova • JQuery • Next.js • LaTeX & Maple • Matlab/Simulink	• Git • Cmake • Jira/Azure • Docker • Figma • Microsoft Office • Linux • Android/iOS	• SolidWorks • Siemens NX • OnShape/AutoCAD • Ansys CFX • Star CCM+ • Abaqus • FreeCAD • Fusion 360	• 3D Printers • Lathe/Mill • Drill Press • Bandsaws • Sanders/Grinders • Oscilloscopes • Soldering • Function Generators

EDUCATION

University of Washington	M.S. Robotics & Data Science • Thesis - 3D Graphics & Regression Models 3.6 • Advisors: Steve Brunton & Ashis Banerjee Python MATLAB C++	Sep 2018 - Dec 2019
Santa Clara University	M.S. Computer Engineering • Half Completed Transferred to University of Washington 3.8 B.S. Mechanical Engineering • Entrepreneurship minor Graduated in 3 years with honors 3.6 MATLAB LaTeX C Simulink LabVIEW Maple Lathe Mill	OpenGL ROS Sep 2017 - Jun 2018 Sep 2014 - Sep 2017

INDUSTRY EXPERIENCE

Pebble Products LLC Founder	• Develop/prototype new hardware/software products • Contract engineering services to local companies/individuals notes.abgup.com	Aug 2022 - Present
Bishop Blanchet High School CS & Robotics Teacher	• Teach high school students engineering skills to design/build a autonomous robot • Teach computer science in Python for Rainier Scholars Java Python	pebbleproducts.com Dec 2020 - Present
Kawasaki Robotics SW Engineer, Robotics	• Developed software to move wafer handling (scara) robots • Tested physical robotics arms on local and vendor facilities to ensure functionality • Optimized for throughput and reach requirements to maximize computer chip production capability AS - Domain Specific Language	OnShape Jun 2022 - Aug 2022
SummerBio SW Engineer, Robotics	• Developed software drivers (6 DOF arms, benchtop systems) for VWorks • Built communication platforms through both Ethernet and serial port protocols • Built hardware testing rigs to ensure new hardware/drivers work with automation line • Unit tested drivers through manual isolation testing and automated via c# test framework, xUnit C# Figma	Oct 2021 - Feb 2022
Lam Research SW/ME Engineer 1y6m	• Led and assisted Android app development via Cordova (JS) framework • Developed/Maintained software to optimize a 6 DOF robotic arm via C# WinForms • Built communication protocols for hardware components via Modbus, Ethernet, and Bluetooth protocols • Developed HMI software and tested HW/SW in a clean room environment via systematic approach • Analyzed test results via data science techniques to ensure repeatability and reliability metrics • Led development of and maintained a fishbone diagram issue diagnosis application with offshore developers via C# Xamarin Framework on Windows and Android • Automated Android content upload and verification via VBA excel automation tools • Worked alongside machinists to develop custom test rigs and hardware components (sheet metal, aluminum/steel parts, injection molded designs, etc) for robotics arm applications • Designed and printed custom 3D components on a MakerBot and local vendors • Designed a custom electronics cart via Siemens NX fitting custom components and facility constraints C# JavaScript Python VBA Figma	OnShape May 2020 - Oct 2021

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[NX](#) || [Teamcenter](#)

CSAA Insurance Physics Consultant 2y	- Evaluated novel engineering and physics aspects for patent applications - Researched alternative approaches for products and methods	Oct 2018 - Oct 2020
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INDUSTRY EXPERIENCE (continued...)

University of Washington Research/Teach Asst. 1y3m	- Taught and led students in graduate mathematics courses for engineering - Developed a custom physics simulation of Optical Tweezers via OpenGL in C++ - Created a digital twin of a compliant motor and optimized model via novel regression techniques.	Sep 2018 - Dec 2019
	C++ Python	OpenGL ROS
Santa Clara University Research/Teach Asst. 1y9m	- Simulated biological membrane fracture through AI methods (<i>see paper</i>) - Assisted teaching course in Numerical Analysis to undergraduate engineering students	Sep 2016 - Jun 2018
	Matlab/LaTeX	

INTERNSHIPS || 2y

Microvision SDE Intern 3m	Modeled the response of Lidar activated SiPM (Silicon photomultipliers)	Summer 2019
	Simulink MATLAB	LTSpice
TheraNova SDE Intern 3m	- Developed a python software analysis system to understand gait measurements - Ensured the software analysis is accurate for 80+ patients	Summer 2018
	Python MATLAB	
Valeo Systems Engr Intern 3m	- Produced hardware and software demos for automotive OEMs - Collaborated with start-ups and OEMs to develop new cabin safety features	Spring 2018
	VBA	
Pentair ME/EE Intern 7m	- Optimized performance of steady state and transient phases of circuit breakers - Laboratory tested and simulated rail heating to melt snow	Winter 2017
	Ansys CFX Solidworks	Oscilloscopes, Function Generators, DC/AC Power Supplies, Soldering
Accel Biotech ME Intern 3m	- Prototyped medical device components and test assemblies - Supported mechanical, electrical, and software design of a blood diagnostic device	Summer 2016
	SolidWorks	Oscilloscopes
Caltrans ME Intern 3m	- Reviewed and advised on structural testing for next generation locomotives - Designed a floor plan using Microsoft Visio & participated in vendor meetings	Summer 2015

PUBLICATIONS, CONFERENCES & PATENTS

Automobile Damage Detection Using Thermal Conductivity		Dec 2018
J. Schow, and A. Gupta (US Patent)		
A Cellular Automaton for Modeling Non-Trivial Biomembrane Ruptures		July 2018
A Gupta, G. Reint, I. Gozen, and M. Taylor		
Presented at <i>The 13th World Congress in Computational Mechanics (WCCM)</i>		Sep 2018
Session: Novel Mathematical Models and Computational Methods, New York, NY		
Published in <i>Soft Matter</i>		
Certifications:	Engineer-In-Training, State of CA	May 2018
	Solidworks CSWA	Mar 2015

VOLUNTEERING ACTIVITIES

Auto Angels, Bellevue Car Mechanic	Jun 2024 - Present
Rainier Scholars, Seattle Computer Science Curriculum TA/Development Assistant Python	Jun 2024 - Present
The Bikery, Seattle , Bicycle Board Member & Technician	Nov 2023 - Present
Capitol Hill Tool Library, Seattle Shop Manager Onshape, Machine Shop Tools (metal/wood/acrylic)	May 2024 - Present
San Jose Bicycle Coalition (SJBC) , Software & Data Consultant	Jul 2020 - Present
Santa Clara University , Alumni Engineering Mentor HTML, CSS, Javascript, Python	Oct 2023 - Present
San Jose Bicycle Clinic , Bicycle Technician	Jul 2020 - Present
FIRST Robotics Competition (FRC) 254, 5940, 1983, & 4180, Engineering Mentor JAVA, Onshape, SolidWorks	Aug 2021 - Present
Reddit Group: r/ControlTheory , Moderator	Dec 2018 - Present

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SOFTWARE PROJECTS

206 Bike Polo

Web Developer

- Update and maintain website using Javascript framework (Next.js)

[HTML](#) || [CSS](#) || [Javascript](#)

[Next.js](#)

[206bikepolo.com](#)

San Jose Bicycle Coalition

Software/Data Consultant

- Use OCR package to automate uploading of sign in sheets into salesforce

[Python](#)

[Salesforce](#)

[bikesiliconvalley.org](#)

Find a Paint

Web Developer

- Developed website to compare paint brands to help high school mom find corresponding paint for work

[Python](#) || [HTML/CSS](#) || [JS](#)

[Flask](#)

[findapaint.com](#)

Masala Blend

Web Developer

- Help stepmother build webpage to sell homemade spices locally

[HTML](#) || [CSS](#) || [Javascript](#)

[Shopify](#) || [Jekyll](#)

[masalablend.com](#)

Personal Website & Resume

Web Developer

- Build personal website via github pages & programmatically render resume via Latex

[HTML](#) || [CSS](#) || [JS](#) || [Latex](#)

[Jekyll](#) || [Xetex](#)

[abgup.com](#)

Peer Porfolio Consulting

Web Developer

- Assist undergraduates build their resume & web portfolio to gain industry experience

[HTML](#) || [CSS](#) || [JS](#) || [Python](#)

[Latex](#) || [Django](#)

[github.com/carly85/personal-resume](#)

Bike Components Database

Web Developer

- Build webpage & database to provide comparison between bicycle components

[Markdown](#)

[github.com/ab12gu/freewheels](#)

HARDWARE PROJECTS

Cap Hill Tool Library

3D Printing Lead

- Assist community members and create models/prints for in-house repairs and personal projects

[OnShape](#)

Bike Polo

Mallet Design/Fabrication

- Model, 3D print, and fabricate custom bike polo mallet

- Iterate through design process, optimizing for weight and reduction of stress concentration/shattering

[Onshape](#)

[Drill](#), [Drill Press](#), [Bandsaw](#)

Custom LEDs

Smart LED Lighting

- Build custom LED timed lighting to match the circadium rhythm of the sun

- Change lighting within building from blue light to gradually hit red light throughout day

[C/C++](#)

[Raspberry Pi](#)

Vehicle Builds and Repair

Repair Tech

- Build custom bicycles in community to encourage human powered transportation

- Repair small and large vehicles in community (Electric Unicycles, Motored vehicles, etc)

[Torque wrenches](#), [hydraulic lifts](#), etc

[Soldering iron](#), [oscilloscopes](#), etc

Bike Component Automation

Automation Lead

- Design and build wireless shifting and cam shafts on bicycles with peers

[FreeCAD](#) || [C/C++](#)

[Arduino Uno](#)

[github.com/ab12gu/bicycle-projects](#)

HOBBIES

Sports

- Juggling
- Handstands
- Bicycling
- Unicycling
- Bike Polo
- Basketball

Sports

- Walking
- Tennis
- Calisthenics
- Running
- Skateboarding
- Rollerskating

Indoor

- Board Games
- Rubix Cube Solving
- Dancing
- Music & Movies
- Drawing
- Painting

Indoor

- Weight Lifting
- Reading
- Video Games
- Cooking
- Yoga
- Socializing

Miscellaneous

- Coding
- Blogging
- Content Creation
- Comedy
- Reddit
- Tinkering

For more: [abgup.com](#)